



10703 - Revealing the Hidden Engines of Little Red Dots: A Statistical Study of BH* Candidates at $6 < z < 9$

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	ebb-mpt	NIRSpec MultiObject Spectroscopy	(1) cosmos-ebb-z7

ABSTRACT

The unique spectral features of Little Red Dots (LRDs) have attracted intense debate regarding their physical nature. The recent discovery of LRDs exhibiting extreme Balmer breaks (EBBs; e.g., MoM-BH-1)* adds further complexity to this emerging population. A theoretical framework, the BH* model, invoking rapidly accreting black holes embedded in dense gas shells has been proposed to explain both the strong Balmer break and the lack of UV emission, offering a compelling interpretation for the elusive V-shaped SEDs of LRDs. However, a statistical spectroscopic sample of LRDs with EBBs is urgently needed to determine whether the BH* scenario represents the dominant mechanism underlying the broader LRD population.

We propose to conduct a NIRSpec/MSA Prism survey of 27 EBB candidates, photometrically selected in the COSMOS field. This program will 1) measure spectroscopic redshifts and confirm the physical nature of these EBB sources; 2) establish the statistical connection between the BH* scenario and the overall abundance of LRDs, providing critical insights into early black hole seeding and growth; 3) deliver the first clustering measurement of EBB systems to constrain their host halo masses and environments; and 4) identify alternative rare populations, including the earliest quiescent galaxies, potential $z > 20$ Lyman-break galaxies, and heavily dust-obscured galaxy/AGN systems.

OBSERVING DESCRIPTION

We will perform 10 NIRSpec/MSA pointings in the COSMOS field to observe our 27 photometrically selected EBB candidates at $6 < z < 9$. To maximize both the number of EBBs observed and the legacy value of this program, we adopt NIRSpec/MSA rather than fixed slit spectroscopy (FSS). We will use the PRISM/CLEAR configuration ($R \sim 100$), which offers broad wavelength coverage and excellent continuum sensitivity, the best observing strategy for the proposed science goals. These observations will capture the rest-frame UV–optical continuum and key diagnostic emission lines such as H-beta and [O III] for sources at $6 < z < 9$, encompassing all major spectral features essential to this program.

For each pointing, we will use 50 groups and one integration per exposure with the NRSIRS2 readout pattern. A standard three-shutter dither pattern will be applied, yielding a total on-source exposure time of 10,985 s (~ 3 hours) per mask. In addition, we will include ~ 1000 filler targets in total, including [O III] emitters, H-alpha emitters, UV-bright high- z galaxies, massive galaxy candidates at $z > 4$, and AGNs across cosmic time—to enhance the legacy value of the dataset.

Proposal 10703 - Targets - Revealing the Hidden Engines of Little Red Dots: A Statistical Study of BH* Candidates at $6 < z < 9$

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	cosmos-ebb-z7	RA: 10 00 7.1928 (150.0299700d) Dec: +02 08 37.91 (2.14386d) Equinox: J2000		
	Comments: Description=[]				

Proposal 10703 - Observation 1 - Revealing the Hidden Engines of Little Red Dots: A Statistical Study of BH* Candidates at $6 < z < 9$

Observation	Proposal 10703, Observation 1: ebb-mpt										Fri Mar 13 19:12:29 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
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	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	4		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	5		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	6		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	7		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	8		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	9		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	10		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold		
	MSATA	false	No	MSA Center	cosmos-ebb-z7 (69 sources)		jwst-nirspec-prism		1.5		

Proposal 10703 - Observation 1 - Revealing the Hidden Engines of Little Red Dots: A Statistical Study of BH* Candidates at $6 < z < 9$

Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	150.14503570833 332 Degrees 1.9564177777777 778 Degrees	240.00466112201 775			3	3	10985.434
	2	1 (PRISM/CLEAR)	c2	3 Shutter Slitlet	149.83018570833 332 Degrees 2.0153119444444 445 Degrees	239.99284816613 013			3	3	10985.434
	3	1 (PRISM/CLEAR)	c3	3 Shutter Slitlet	150.00766533333 334 Degrees 2.108095 Degrees	239.99924946792 015			3	3	10985.434
	4	1 (PRISM/CLEAR)	c4	3 Shutter Slitlet	150.029067 Degrees 2.218272222222 22 Degrees	239.99982444173 685			3	3	10985.434
	5	1 (PRISM/CLEAR)	c5	3 Shutter Slitlet	150.19312316666 665 Degrees 2.2343344444444 444 Degrees	240.00583101645 82			3	3	10985.434
	6	1 (PRISM/CLEAR)	c6	3 Shutter Slitlet	150.24613241666 665 Degrees 2.295382222222 22 Degrees	240.00767128851 587			3	3	10985.434
	7	1 (PRISM/CLEAR)	c7	3 Shutter Slitlet	150.09066770833 33 Degrees 2.167209166666 664 Degrees	240.00219379759 1			3	3	10985.434
	8	1 (PRISM/CLEAR)	c8	3 Shutter Slitlet	150.09040654166 665 Degrees 2.1093244444444 443 Degrees	240.00230072826 24			3	3	10985.434
	9	1 (PRISM/CLEAR)	c9	3 Shutter Slitlet	149.82808879166 666 Degrees 2.246664166666 667 Degrees	239.99225632159 244			3	3	10985.434
	10	1 (PRISM/CLEAR)	c10	3 Shutter Slitlet	149.9275505 Degrees 2.021774722222 22 Degrees	239.99647532219 453			3	3	10985.434

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Special Requirements

Group Visits within 53.0 Days
Aperture PA Range 238 to 242 Degrees (V3 99.4254303 to 103.4254303)
Visits Same PA
MSA Planned Aperture PA 240.0000 to 240.0000 Degrees (V3 101.4254303 to 101.4254303)